ISO/IEC 25010 System and Software Quality Models

Quality Characteristics	Quality Sub-Characteristics
Functional Suitability: A set of attributes that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	**1. Functional completeness:** **Definition:** The degree to which the set of functions covers all the specified tasks and intended users' objectives. **2. Functional correctness:** **Definition:** The degree to which a product or system provides accurate results when used by intended users. **Example:** The system shall accurately calculate Value Added Tax (VAT) based on the 25% tax. 3. **Functional appropriateness:** **Definition:** The degree to which the functions facilitate the accomplishment of specified tasks and objectives. **Example:** The system should allow users to register by providing their name, email, and password.
Performance Efficiency: A set of attributes represents the degree to which a product performs its functions within specified time and throughput parameters and is efficient in using resources (such as CPU, memory, storage, network devices, energy, materials...) under specified conditions.	**1. Time behaviour** **Definition:** The degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements. **Example:** The system must achieve an average response time of less than 2 seconds for executing a search query. **2. Resource utilization** **Definition:** The degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements. **Example:**

	The system must limit the application's memory usage to no more than 500 MB during normal operation. **3. <u>Capacity</u>** **Definition:** The degree to which the maximum limits of a product or system parameter meet requirements. **Example:** The system should have a maximum response time of 500 milliseconds for 95% of user interactions under normal load conditions.
<u>Compatibility</u> A set of attributes that represents the degree to which a product, system, or component can exchange information with other products, systems, or components and/or perform its required functions while sharing the same common environment and resources.	**1. <u>Co-existence</u>** **Definition:** The degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products without detrimental impact on any other product. **Example:** The system should function correctly on major browsers: Chrome, Firefox, Safari, and Edge. **2. <u>Interoperability</u>** **Definition:** The degree to which a system, product, or component can exchange information with other products and mutually use the information that has been exchanged. **Example:** The system should support JSON and XML data formats for data exchange.
Interaction Capability (Usability) A set of attributes that represents the degree to which a product or system can be interacted with by specified users to exchange information is the user interface to complete specific tasks in a variety of contexts of use.	**1. <u>Appropriateness Recognizability</u>** **Definition:** The degree to which users can recognize whether a product or system is appropriate for their needs. **Example:** The user interface must be designed with **clear and intuitive labels**, icons, and navigation elements.

	2. Learnability **Definition:** The degree to which the functions of a product or system can be learned to be used by specified users within a specified amount of time. **Example:** The application should have context-sensitive help accessible via tooltips or a dedicated help section. **3. Operability** **Definition:** The degree to which a product or system has attributes that make it easy to operate and control. **Example:** The system should implement Role-Based Access Control to manage user permissions based on roles **4. User Error Protection** **Definition:** The degree to which a system prevents users from operation errors. **Example:** The system must display a confirmation dialog to prevent accidental clicks for critical functions like deleting an account. **5. User Engagement** **Definition:** The degree to which a user interface presents functions and information in an inviting and motivating manner, encouraging continued interaction. **Example:** The system should provide personalized content recommendations based on user preferences, browsing history, or past interactions. **6. Inclusivity** **Definition:** The degree to which a product or system can be used by people of various backgrounds (such as people of various ages, abilities,

	cultures, ethnicities, languages, genders, economic situations, etc.). **Example:** The system must support voice recognition-based navigation for users who cannot use a mouse or touch screen. **7. <u>User assistance</u>** **Definition:** The degree to which a product can be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use. **Example:** The system must provide contextual help features that are accessible from within the user interface to assist users in performing tasks and resolving issues. **8. <u>Self-descriptiveness</u>** **Definition:** The degree to which a product presents appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources (such as user documentation, help desks, or other users). **Example:** The system should display contextual hints explaining their function when users hover over UI elements
<u>Reliability</u> A set of attributes represents the degree to which a system, product, or component performs specified functions under specified conditions for a specified period.	**1. <u>Faultlessness</u>** **Definition:** The degree to which a system, product, or component performs specified functions without fault under normal operation. **Example:** The system must achieve 99% reliability during system start-up after 600 on-off cycles of operation.

	2. <u>Availability</u> **Definition:** The degree to which a system, product, or component is operational and accessible when required for use. **Example:** The payment processing system must achieve an availability of at least 99.9% during business hours (e.g., 9:00 AM to 5:00 PM) on weekdays. **3. <u>Fault Tolerance</u>** **Definition:** The degree to which a system, product, or component operates as intended despite hardware or software faults. **Example:** The software application must be deployed on a backup server infrastructure that includes at least two geographically distributed servers. **4. <u>Recoverability</u>** **Definition:** The degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the system's desired state. **Example:** The system must implement regular automated backups of critical data and configuration settings to a secure off-site location
<u>Security</u> The degree to which a product or system defends against attack patterns by malicious actors and protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	**1. <u>Confidentiality</u>** **Definition:** The degree to which a product or system ensures that data are accessible only to those authorized. **Example:** The system must implement secure user authentication mechanisms to ensure only authorized users can access sensitive data and perform privileged actions. **2. <u>Integrity</u>** **Definition:**

	The degree to which a system, product, or component ensures that the state of its system and data are protected from unauthorized modification or deletion, either by malicious action or computer error. **Example:** The system must implement comprehensive data validation mechanisms to ensure the integrity of user input data and prevent malicious input or tampering. **3. <u>Non-repudiation</u>** **Definition:** The degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later. **Example:** The system must support digital signatures that adhere to industry-standard RSA cryptographic protocols for sensitive transactions or communications. **4. <u>Accountability</u>** **Definition:** The degree to which the actions of an entity can be traced uniquely to the entity. **Example:** The system must implement audit logging mechanisms to record all user actions, system events, and security-related activities. **5. <u>Authenticity</u>** **Definition:** The degree to which the identity of a subject or resource can be proved to be the one claimed. **Example:** The system must implement multi-factor user authentication to verify the authenticity of users accessing the system **6. <u>Resistance</u>** **Definition:** The degree to which the product or system sustains operations while under attack from a malicious actor.

	Example The system shall have a proactive security patch management process in place to promptly identify, assess, and apply security patches and updates for all software components and dependencies.
Maintainability This characteristic represents the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct, or adapt to changes in the environment and requirements.	**1. <u>Modularity</u>** **Definition:** The degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components. **Example:** The system shall enable features to be added, updated, or removed during runtime without requiring downtime or interruption to ongoing operations **2. <u>Reusability</u>** **Definition:** The degree to which a product can be used as an asset in more than one system or in building other assets. **Example:** The system shall provide a database loader that supports any ODBC database **3. <u>Analyzability</u>** **Definition:** Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified. **Example:** The system shall provide a debugging mode that logs the details for all component invocations. **4. <u>Modifiability</u>** **Definition:** The degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.

	Example: The Medication Monitoring System must provide a configurable alerting mechanism that allows healthcare providers to easily modify and customize medication alert thresholds and parameters based on patient-specific factors and clinical guidelines. **5. <u>Testability</u>** **Definition:** Degree of effectiveness and efficiency with which test criteria can be established for a system, product, or component, and tests can be performed to determine whether those criteria have been met.
Flexibility The degree to which a product can be adapted to changes in its requirements, contexts of use, or system environment.	**1. <u>Adaptability</u>** **Definition:** The degree to which a product or system can effectively and efficiently be adapted for or transferred to different hardware, software, or other operational or usage environments. **Example:** The system shall be able to execute on-premises or in a cloud-hosted environment. **2. <u>Scalability</u>** **Definition:** The degree to which a product can handle growing or shrinking workloads or adapt its capacity to handle variability. **Example:** The UTM Website shall be scalable to accommodate unrestricted growth in students taking on-demand courses. **3. <u>Installability</u>** **Definition:** Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment. **Example:** Installation packages should be provided in platform-neutral formats, such as executable installers or compressed archives, to facilitate deployment on different operating systems.

	4. <u>Replaceability</u> **Definition:** The degree to which a product can replace another specified software product for the same purpose in the same environment. **Example:** The software application must support the replaceability of third-party components, allowing users or administrators to easily substitute or upgrade external libraries, frameworks, or services used by the application.
Safety This characteristic represents the degree to which a product under defined conditions avoids a state in which human life, health, property, or the environment is endangered.	**1. <u>Operational constraint</u>** **Definition:** When encountering operational hazards, the degree to which a product or system constrains its operation within safe parameters or states. **Example** The software-controlled system must include an emergency stop function that immediately halts all system operations in case of an emergency **2. <u>Risk identification</u>** **Definition:** The degree to which a product can identify a course of events or operations that can expose life, property, or environment to unacceptable risk. **Example** The Medication Monitoring System must incorporate automatic detection mechanisms to identify potential adverse drug events (ADEs) or medication errors based on patient medication data and clinical indicators. **3. <u>Fail-Safe</u>** **Definition:** The degree to which a product can automatically place itself in a safe operating mode or revert to a safe condition in case of failure. **Example:** The Medication Monitoring System shall include automatic dosing limitation

	mechanisms to prevent patients from receiving medication doses that exceed safe thresholds. **4. <u>Hazard warning</u>** **Definition:** The degree to which a product or system provides warnings of unacceptable risks to operations or internal controls so that they can react in sufficient time to sustain safe operations. **Example:** The Medication Monitoring System must provide real-time drug interaction alerts to healthcare providers when potential interactions between prescribed medications are detected. **5. <u>Safe integration</u>** **Definition:** The degree to which a product can maintain safety during and after integration with one or more components. **Example:** The medication monitoring system must be securely integrated with electronic health record (EHR) systems to exchange patient medication data and ensure interoperability with existing healthcare information systems.